Steering Vector Decay: Measuring the Rapid Attenuation of Activation Steering in Transformer Hidden States

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Abstract

Activation steering—adding a learned direction to a language model's hidden states at inference time—is among the simplest methods for controlling model behavior without fine-tuning. Practitioners must choose whether to steer continuously (risking degenerate output) or for a limited window, yet the actual dynamics of how steering influence persists after the vector is removed have never been directly measured. We present the first systematic characterization of steering vector decay in transformer hidden states. Using contrastive activation addition for sycophancy on QWEN2.5-1.5B-INSTRUCT (28 layers, 1.5B parameters), we apply the steering vector for $N \in \{1, 3, 5, 10\}$ initial tokens, then measure the signed projection of the hidden-state perturbation onto the steering direction at each subsequent step under teacher-forced generation. We find that steering influence decays to less than 5% of its peak within a single token after the vector is removed, following an exponential with time constants $\tau \approx 1.2$–5.8 tokens (R^2 up to 0.97). The KV cache is the primary vehicle for any residual signal: preserving it retains a small positive trace, while resetting it produces a statistically significant negative rebound (overcorrection; Cohen's d up to 3.41, $p < 0.0001$). These results demonstrate that continuous steering is necessary because transformer dynamics aggressively wash out transient perturbations, and they provide mechanistic support for KV-cache-based steering as a more persistent alternative.

1 Introduction

Activation steering lets practitioners control language model behavior at inference time by adding a learned direction to the residual stream [Turner et al., 2023, Rimsky et al., 2024]. The approach is appealingly simple: compute a steering vector from contrastive examples, add it with a scalar coefficient α, and the model shifts toward the desired behavior. But how long does this shift last if the vector is only applied for the first few generation steps?

Prior work offers a clear qualitative answer—initial-only steering fails [Scalena et al., 2024]—but no quantitative one. Scalena et al. [2024] show that steering only the first generated token produces text that quickly reverts to baseline behavior. Xie [2025] propose a heuristic decay formula $\alpha_t = \alpha \cdot (1/(1 + \omega t))^k$ to gradually reduce steering strength, but this formula describes a *desired* decay schedule, not the *natural* decay of the steering signal in hidden states. The fundamental question remains unanswered: when a steering vector is removed, how quickly does its influence actually leave the model's representations?

This question matters for engineering and for understanding transformer dynamics. If the steering signal persists for many tokens, practitioners could save compute by steering only early steps. If it vanishes instantly, continuous steering is unavoidable—or alternative persistence mechanisms (such as KV cache modification) become necessary. More broadly, the decay rate reveals how robustly transformers correct transient perturbations to their residual stream, with implications for adversarial robustness and mechanistic interpretability.

We present the first direct measurement of steering vector decay curves. Our approach isolates the decay mechanism by using *teacher-forced generation*: we first generate tokens under continuous steering, then replay the same token sequence while steering only the first N steps. By holding token identity constant, we separate the steering signal in hidden states from confounds introduced by different token sequences. We measure the signed projection of the hidden-state perturbation onto the steering direction at each generation step, capturing both the magnitude and direction of residual influence.

Our main findings on QWEN2.5-1.5B-INSTRUCT with sycophancy steering are:

- The steering signal decays to less than 5% of its peak within one token after the vector is removed, with exponential time constants $\tau \approx 1.2$–5.8 tokens.
- The KV cache is the primary persistence mechanism: preserving it retains a small positive residual, while resetting it causes a negative rebound (overcorrection; Cohen's d up to 3.41).
- Counterintuitively, more initial steering tokens produce *faster* relative decay (shorter τ), suggesting stronger corrective dynamics in response to larger perturbations.

The rest of this paper is organized as follows. Section 2 reviews activation steering and related work on steering schedules. Section 3 describes our experimental methodology. Section 4 presents the decay curves, exponential fits, and KV cache analysis. Section 5 interprets these findings and discusses limitations. Section 6 summarizes our contributions.

2 Related Work

Activation steering. Turner et al. [2023] introduce Activation Addition (ACTADD), which steers language models by adding the difference in activations between a pair of contrastive prompts to the residual stream. Rimsky et al. [2024] scale this to Contrastive Activation Addition (CAA), averaging activation differences across hundreds of contrastive pairs and applying the resulting vector at every generated token position. Li et al. [2023] propose Inference-Time Intervention (ITI), which shifts activations at a limited number of attention heads to improve truthfulness. All three methods apply steering continuously at every forward pass, implicitly assuming that one-shot steering would be insufficient—an assumption our work directly tests and confirms.

Steering schedules and decay. Scalena et al. [2024] compare multiple steering schedules: start-only (first step), fixed (constant α), diminishing (linear decay), and dynamic (KL-divergence-adaptive). They show that start-only steering fails to maintain behavioral conditioning, while their dynamic method reveals that the optimal steering intensity drops sharply after the first few tokens. Xie [2025] propose a token-wise decaying strategy $\alpha_t = \alpha \cdot (1/(1+\omega t))^k$ for SAE-based steering, designed to prevent degeneration from constant high-strength intervention. Klerings et al. [2025] systematically study steering strength, location, and duration for grammatical tense, finding that late or extended steering causes topic shift and degeneration. These works prescribe how steering *should* decay but do not measure how it *naturally* decays in hidden states after removal. Our work fills this gap with direct measurements.

Representation geometry of steered behaviors. Arditi et al. [2024] show that refusal in language models is mediated by a single direction in the residual stream, demonstrating that some behaviors have strong, persistent linear representations. Tan et al. [2024] evaluate the generalization and reliability of steering vectors, finding substantial variability across inputs and spurious biases. Subramani et al. [2022] demonstrate that steering vectors can precisely control generation when applied correctly. Our decay measurements complement this line of work by characterizing the *temporal* persistence of steered representations across generation steps.

KV cache steering. Belitsky et al. [2025] propose one-shot steering via KV cache modification, where cached key-value pairs are modified once before generation and persist through the attention mechanism. They contrast this with activation steering, which requires continuous intervention, implicitly demonstrating that activation steering decays while KV cache modifications persist. Our results provide direct evidence for this contrast: we show that the KV cache is the primary vehicle for any residual steering signal after activation-based intervention stops, and that removing steered KV pairs causes the model to overcorrect.

3 Methodology

3.1 Overview

We isolate the decay of steering influence by comparing hidden states under different steering conditions while controlling for token identity via teacher-forcing. The key design choice is using *incremental generation with KV cache*, which properly captures how steered key-value pairs from earlier steps persist in attention and influence subsequent hidden states.

3.2 Model and Steering Vector

Model. We use QWEN2.5-1.5B-INSTRUCT [Qwen Team, 2024], a 28-layer, 1.5B-parameter instruction-tuned model. We select this model for ungated access, fast iteration, and meaningful instruction-following behavior.

Steering vector extraction. We compute a sycophancy steering vector using the CAA dataset from Rimsky et al. [2024], which contains question-answer pairs where one answer matches a persona's implied preference (sycophantic) and the other does not. We extract mean activation differences across 300 contrastive pairs at layers $\{0, 7, 14, 21, 28\}$. The resulting vector norms vary substantially: layers 0 and 7 produce near-zero vectors ($\|\boldsymbol{s}\| < 0.3$), while upper layers yield much stronger signals (layer 21: $\|\boldsymbol{s}\| = 3.66$; layer 28: $\|\boldsymbol{s}\| = 5.82$). We steer at **layer 21**, where the vector is strong and upstream of the measurement point.

Steering application. We add the steering vector to the residual stream output at layer 21 with coefficient $\alpha = 8.0$, using a forward hook that intercepts the layer output during the forward pass.

3.3 Experimental Conditions

For each of 60 held-out evaluation prompts, we first generate 40 tokens under **continuous steering** (steering at every step) using greedy decoding, producing a reference token sequence $T = (t_1, t_2, \ldots, t_{40})$. We then teacher-force T under the following conditions:

- **Baseline**: Teacher-force T with no steering. The KV cache accumulates naturally.
- **Continuous**: Teacher-force T with steering at every step.
- **Initial-N**: Teacher-force T with steering for the first N steps only. The KV cache is preserved, so steered key-value pairs from steps $1, \ldots, N$ remain in the cache during steps $N + 1, \ldots, 40$.
- **Initial-N-NoCache**: Same as Initial-N, but the KV cache is *reset* at step N. This removes all steered key-value pairs, isolating the effect of the KV cache.
- **Free-N**: Free (autoregressive) generation with steering for the first N steps. No teacher-forcing; the model generates its own continuations.

We test $N \in \{1, 3, 5, 10\}$ for a total of 12 conditions per prompt (1 baseline + 1 continuous + 4 initial + 4 no-cache + 4 free, though we focus analysis on the first 10 teacher-forced conditions).

3.4 Measurement

At each generation step, we capture the hidden state $\boldsymbol{h}_t$ at **layer 28** (the final layer, downstream of the steering intervention at layer 21) using a forward hook. We compute the perturbation:

$$\Delta_{\text{proj}}(t) = \frac{(\boldsymbol{h}_t^{\text{steered}} - \boldsymbol{h}_t^{\text{baseline}}) \cdot \hat{\boldsymbol{s}}}{\|\hat{\boldsymbol{s}}\|} \tag{1}$$

where $\hat{\boldsymbol{s}}$ is the unit-normalized steering vector at layer 21. This *signed projection* quantifies how much of the hidden-state perturbation remains aligned with the steering direction: positive values indicate residual steering influence, negative values indicate overcorrection, and zero indicates no effect.

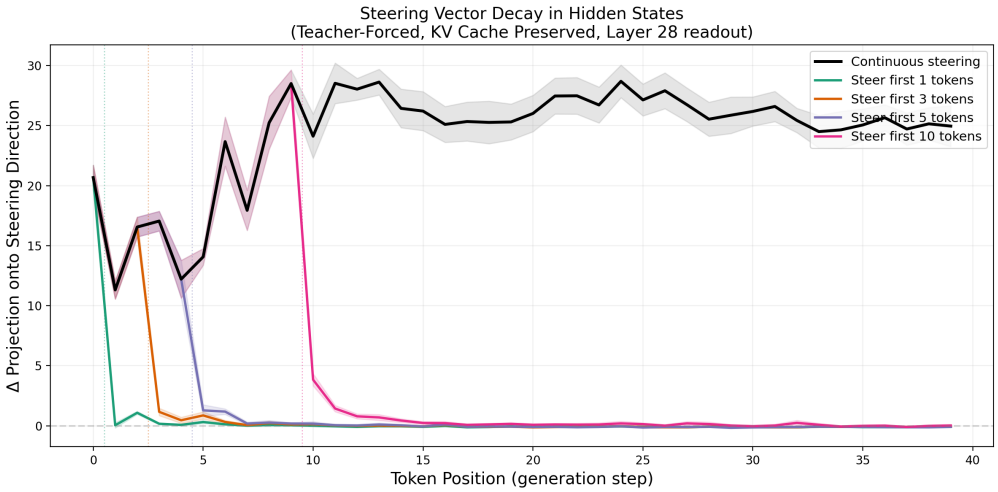

Figure 1: Steering influence (Δ_{proj}) over generation steps. The steering vector is applied for the first N steps (shaded region), then removed. Under all initial-steering conditions (with KV cache preserved), the perturbation drops to near-zero within 1–2 tokens after removal. Continuous steering (gray dashed) maintains $\Delta_{\text{proj}} \approx 24$ throughout.

3.5 Curve Fitting and Statistical Analysis

We fit an exponential decay model to the post-steering Δ_{proj} values:

$$\Delta_{\text{proj}}(t) = A \cdot \exp\left(-\frac{t-N}{\tau}\right) + C \tag{2}$$

where A is the amplitude, τ is the time constant (in tokens), C is the asymptotic residual, and $t - N$ indexes steps after the steering stops. We report τ, R^2 (goodness of fit), and derived metrics including half-life and time to 10% of peak.

To test the KV cache effect, we use paired t-tests comparing Δ_{proj} values (averaged over 10 post-steering steps) between the Initial-N and Initial-N-NoCache conditions across all 60 prompts, reporting Cohen's d for effect size.

3.6 Reproducibility

All experiments use random seed 42, greedy decoding (temperature 0), Python 3.12.8, PyTorch 2.10.0, and Transformers 5.3.0. Prompts are formatted in ChatML format. We run on a single NVIDIA RTX A6000 GPU (49GB). The full experiment (60 prompts $\times$ 12 conditions) completes in approximately 16 minutes.

4 Results

4.1 Steering Influence Decays Within One Token

Figure 1 shows the Δ_{proj} curves for all steering conditions. During the steered window (shaded), the projection reaches values of 12–29 depending on condition. The moment the steering vector is removed, Δ_{proj} drops precipitously. Within one token, the signal falls to less than 5% of its peak value across all conditions.

Table 1 quantifies this decay. The post-steering residual (averaged over 5 steps after removal) ranges from 1.7% of peak ($N = 1$) to 5.1% ($N = 5, 10$). Even the largest residual (1.45 for $N = 10$) is small compared to the continuous-steering mean of 24.31.

Table 1: Steering decay statistics across initial-steering conditions (teacher-forced, KV cache preserved). Peak Δ_{proj} is the maximum value during steering. Post-5 and Post-10 averages are computed over the 5 and 10 steps immediately after steering stops. Best (slowest) decay highlighted in **bold**.

N	Peak Δ_{proj}	Post-5 avg	Post-10 avg	% of peak	τ (tokens)	R^2
1	20.67	0.34	0.20	1.7%	**5.82**	0.506
3	16.55	0.58	0.34	3.5%	3.35	0.889
5	12.21	0.63	0.36	**5.1%**	2.42	0.905
10	28.50	**1.45**	**0.81**	**5.1%**	1.21	**0.974**
Continuous	24.31 (mean)	—	—	100%	—	—

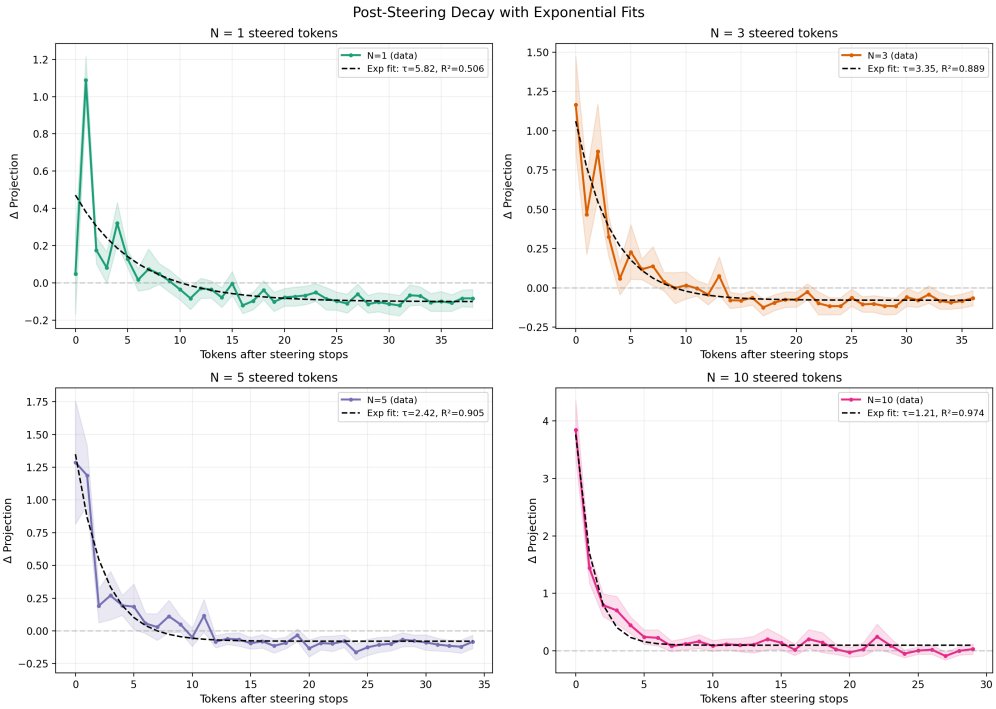

Figure 2: Exponential decay fits (Eq. 2) to post-steering Δ_{proj} values. Solid lines show observed means across 60 prompts; dashed lines show fitted curves. The time constant τ decreases with more initial steering tokens, indicating that larger perturbations trigger faster corrective dynamics.

4.2 Exponential Decay Fits

Figure 2 shows the exponential fits to post-steering Δ_{proj} values. The fit quality improves with more steered tokens: $R^2 = 0.51$ for $N = 1$ (barely any post-steering signal to fit) and $R^2 = 0.97$ for $N = 10$. The time constants decrease with N: $\tau = 5.82$ tokens for $N = 1$ down to $\tau = 1.21$ tokens for $N = 10$.

This inverse relationship between N and τ is counterintuitive: one might expect more steering to produce slower decay, but instead the model corrects faster. In absolute terms, $N = 10$ does produce a higher residual (1.45 vs. 0.34 for $N = 1$), but relative to its much larger peak, the decay is faster.

4.3 KV Cache Is the Primary Persistence Mechanism

Figure 3 compares the Initial-N condition (KV cache preserved) with the Initial-N-NoCache condition (KV cache reset after steering). The results reveal a striking asymmetry:

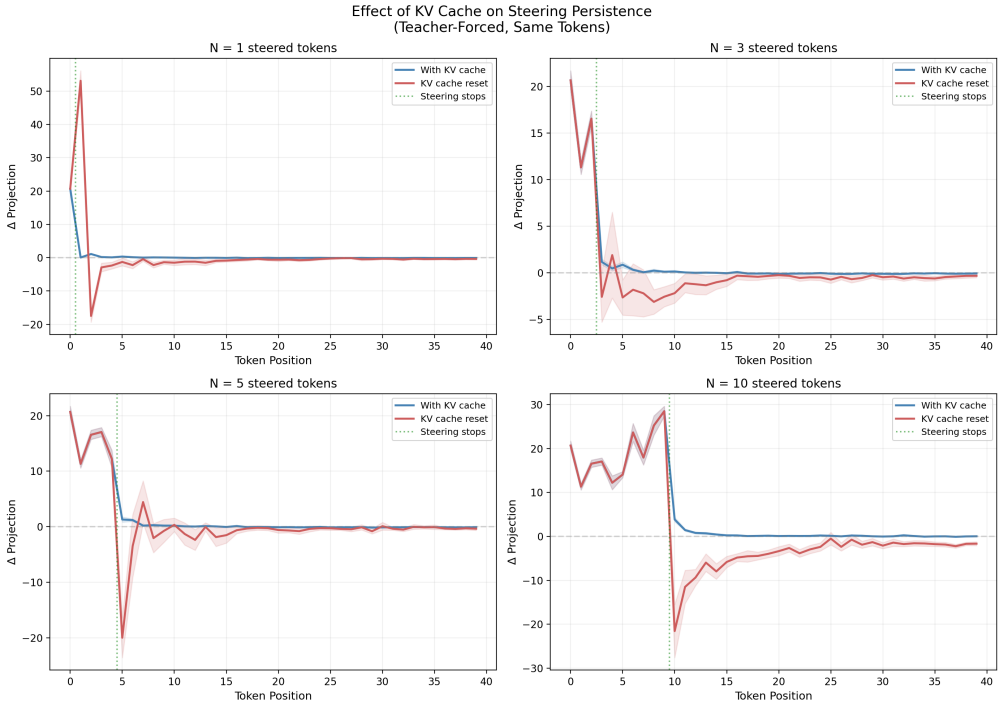

Figure 3: KV cache effect on steering persistence. *(Top)*: Initial-N with KV cache preserved shows a small positive residual after steering stops. *(Bottom)*: Resetting the KV cache at step N produces a negative rebound (overcorrection), indicating the model adapts to the presence of steered attention patterns.

Table 2: Paired t-test comparing post-steering Δ_{proj} (averaged over 10 steps) between KV-cache-preserved and KV-cache-reset conditions across 60 prompts. Positive Cohen's d indicates the KV cache preserves a more positive (steering-aligned) residual.

N	t-statistic	p-value	Cohen's d	Interpretation
1	-5.22	< 0.0001	-0.98	Cache reset $\Rightarrow$ overcorrection
3	2.75	0.008	0.53	KV cache preserves positive residual
5	3.74	0.0004	0.74	Strong KV cache effect
10	17.83	< 0.0001	**3.41**	Very large KV cache effect

- **With KV cache**: Δ_{proj} stays slightly positive after steering stops, indicating a small residual alignment with the steering direction.
- **Without KV cache**: Δ_{proj} goes *negative*, meaning the model's hidden states are pushed *away* from the steering direction relative to baseline—an overcorrection.

Table 2 reports the paired t-test results. The difference is statistically significant for all N values ($p < 0.01$), with effect sizes ranging from moderate ($d = 0.53$ for $N = 3$) to very large ($d = 3.41$ for $N = 10$). The negative t-statistic for $N = 1$ indicates that even a single steered KV pair, when removed, causes the model to overcorrect more strongly than with the cache preserved.

4.4 Normalized Decay Curves

Figure 4 shows all conditions normalized to their peak value. Regardless of N, all curves drop below 10% of peak within 1–3 tokens after the steering vector is removed. This confirms that the rapid decay is a robust property of the system, not an artifact of particular N values or absolute magnitudes.

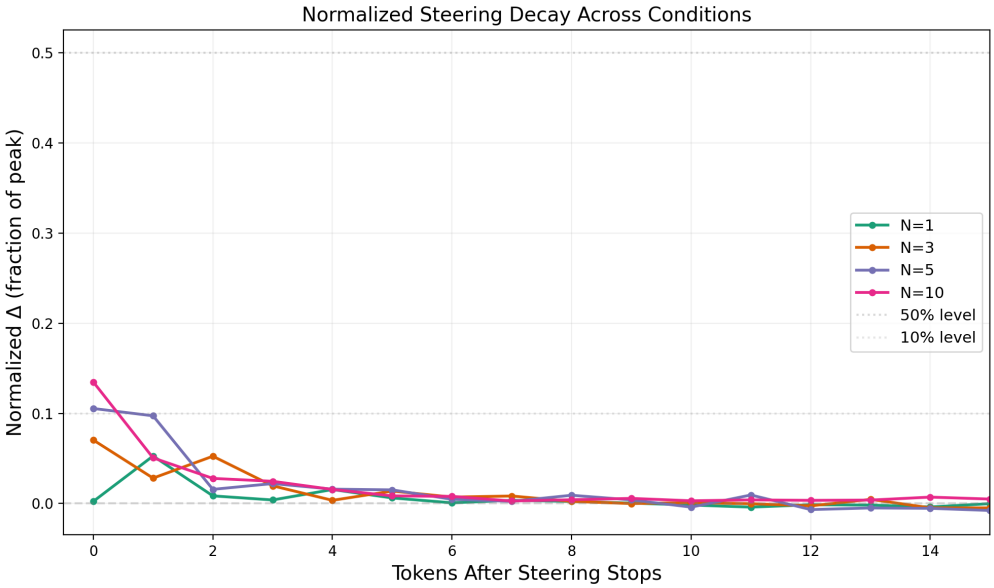

Figure 4: Normalized decay curves ($\Delta_{\text{proj}}/\Delta_{\text{proj}}^{\text{peak}}$) for all initial-steering conditions. All curves drop below 10% within 1–3 tokens after the steering vector is removed, confirming that near-instantaneous decay is robust across steering durations.

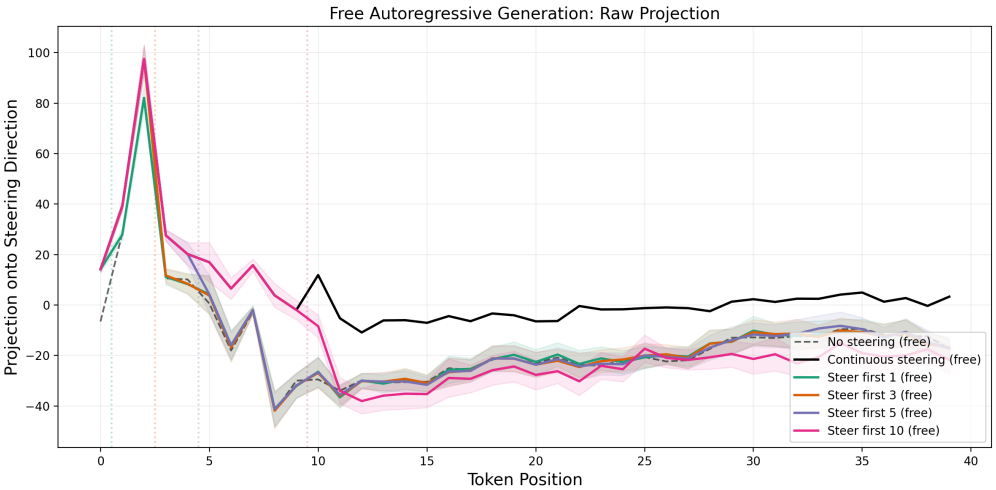

Figure 5: Steering projection under free autoregressive generation (no teacher-forcing). The signal is substantially noisier than teacher-forced conditions because token sequences diverge from baseline, producing different activation patterns at each step. The general trend of rapid decay is consistent with teacher-forced results.

4.5 Free Generation

Figure 5 shows the Δ_{proj} curves under free (autoregressive) generation, where the model generates its own tokens after the initial steering window. These curves are substantially noisier than the teacher-forced conditions because different tokens at each step produce different activation patterns. The general trend—rapid decay followed by oscillation near zero—is consistent with the teacher-forced results, but the noise makes precise characterization difficult. This highlights why the controlled, teacher-forced experimental design is necessary for measuring decay dynamics.

5 Discussion

Transformers aggressively correct transient perturbations. The most striking finding is the speed of decay: the steering signal drops below 5% of its peak within a single token. This is far faster than the decay schedules proposed by Xie [2025], whose formula $\alpha_t = \alpha \cdot (1/(1 + \omega t))^k$ with $k = 3$–4 reaches near-zero only after 8–9 steps. Our results show that the *natural* decay of a perturbation in the residual stream is nearly instantaneous, suggesting that the transformer's layered computation acts as a strong low-pass filter on transient signals. Each forward pass through the remaining layers (22–28 in our setup) effectively washes out the perturbation from the residual stream.

The KV cache as a persistence mechanism. Our KV cache analysis explains *why* some prior work finds that initial steering has any effect at all. When steered key-value pairs remain in the cache, subsequent tokens attend to them, preserving a small positive trace of the steering direction in the hidden states. When these pairs are removed, the model overcorrects—hidden states move *away* from the steering direction relative to baseline. This overcorrection suggests that the model partially adapts its computation to compensate for the steered attention patterns, and removing them leaves the compensatory adjustments without the signal they were compensating for.

This finding directly supports Belitsky et al. [2025], who propose KV cache modification as a one-shot steering method precisely because cached representations persist through attention. Our results provide the mechanistic explanation: activation steering decays because the residual stream is recomputed at each step, while KV cache entries are fixed once written and influence all subsequent steps through attention.

More steering produces faster relative decay. Counterintuitively, the time constant τ decreases from 5.82 tokens ($N = 1$) to 1.21 tokens ($N = 10$). One possible explanation is a nonlinear corrective mechanism: larger perturbations to the residual stream trigger stronger self-correction in subsequent layers, perhaps through attention heads that act as "error-correcting" circuits [Arditi et al., 2024]. Alternatively, with more steered KV pairs, the transition from steered to unsteered computation at step N creates a sharper discontinuity that the model resolves more quickly.

Implications for steering practice. Our results have direct practical consequences. First, initial-only steering is ineffective: practitioners should not rely on steering a few early tokens and expecting the effect to persist. Second, continuous steering is necessary but should use a gradual decay schedule [Scalena et al., 2024, Xie, 2025] to prevent degeneration. Third, for applications requiring persistent behavioral modification from a single intervention, KV cache steering [Belitsky et al., 2025] is mechanistically better suited than activation addition.

Limitations. Our study has several limitations that constrain the generality of our conclusions:

- **Single model**: We study QWEN2.5-1.5B-INSTRUCT only. Larger models with more attention heads and layers may maintain the KV cache signal for longer, or may exhibit different decay dynamics.
- **Single behavior**: Sycophancy may not be representative. Behaviors with stronger linear representations (e.g., refusal [Arditi et al., 2024]) might decay differently.
- **Single-layer intervention**: We steer at layer 21 only. Multi-layer steering or intervention at different layers could produce different persistence characteristics.
- **Single measurement layer**: We measure Δ_{proj} at layer 28. The full picture of how perturbations propagate through intermediate layers is not captured.
- **Model scale**: At 1.5B parameters, our model is relatively small. Scaling laws for steering decay remain unknown.

6 Conclusion

We present the first direct measurement of how steering vector influence decays in transformer hidden states after the vector is removed. Using teacher-forced generation to control for token identity, we show that the steering signal drops to less than 5% of its peak within a single token, following an exponential decay with time constants $\tau \approx 1.2$–5.8 tokens. The KV cache is the primary vehicle for any residual steering signal: preserving it retains a small positive trace, while resetting it causes a statistically significant overcorrection (Cohen's d up to 3.41). These findings

confirm that continuous steering is necessary for sustained behavioral control via activation addition, and provide mechanistic support for KV-cache-based methods as a more persistent alternative.

Several directions for future work emerge. First, repeating this analysis across model scales (7B, 70B+) would test whether decay rate is scale-dependent. Second, comparing decay across different steered behaviors (e.g., refusal, language, reasoning style) would reveal whether some behaviors produce more persistent perturbations. Third, the KV cache overcorrection phenomenon warrants further investigation: does it have downstream behavioral consequences, and can it be exploited for steering in the opposite direction? Finally, measuring decay under multi-layer steering would test whether distributing the perturbation across layers increases persistence.

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